Department of Electrical Engineering
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*Student Capstone Project Proposal • Academic Year 114*

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# AURA

## A Hybrid CNN–Transformer Architecture for 3D Multimodal Brain Tumor Segmentation

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## 1 Project Abstract

This project proposes **AURA** (*Attention · U-Net · Residual · All-modality*), a 3D hybrid CNN–Transformer model for brain tumor segmentation on multi-modal MRI. AURA combines a residual U-Net with a single Transformer bottleneck, integrates uncertainty estimation and calibration so the model can signal when it is likely wrong, and is delivered as a browser-based clinical workstation deployable on a single consumer GPU.

## 2 Background and Motivation

Brain tumors are highly lethal, with a five-year survival rate of roughly 30%. MRI is the diagnostic standard, but manually contouring the three nested tumor regions—enhancing tumor (ET), tumor core (TC), and whole tumor (WT)—across four modalities takes hours per case and varies between readers. Automation is hard: tumor voxels make up only ∼1% of a scan and tumor location varies widely.

U-Nets remain the leading family on BraTS, while Transformer hybrids have recently shown competitive global-context modeling. Full 3D Transformers are heavy and data-hungry, and few methods treat **uncertainty and calibration** as first-class outputs despite their importance for clinical use.

**Objectives.** (1) place self-attention *only* at the deepest bottleneck for an accuracy/cost balance; (2) produce calibrated voxel-level uncertainty so low-confidence cases can be flagged; (3) compare AURA against a plain U-Net and a heavier multi-module variant under one identical recipe; (4) deliver a browser-based clinical workstation.

## 3 Methods and Procedures

**Dataset.** BraTS 2021 (1,251 cases, four modalities), split 1,000 / 251 for training and validation; Z-score normalized per modality on brain voxels.

**Model.** A 3D U-Net with residual CNN encoder/decoder and a Transformer block at the deepest stage. Convolutions handle local texture; bottleneck attention adds global context. Decoder dropout doubles as the source for MC Dropout. Deep-supervision heads are used during training only.

**Training.** Region-wise Dice + Focal loss on the three evaluation regions, so the objective matches the metric. Patch-based training ($128^3$) with tumor-centered sampling addresses class imbalance. Standard practice for the rest: AdamW, cosine schedule with warm-up, mixed precision, EMA, gradient clipping. All development runs on a single consumer GPU.

**Evaluation.** Sliding-window inference, 8-way flip TTA, and connected-component post-processing. Models are assessed along three axes: *accuracy* (Dice, HD95), *calibration* (ECE), and *uncertainty quality* (AURC, selective prediction). Pairwise comparisons use the paired Wilcoxon signed-rank test with bootstrap 95% CIs.

**Comparison.** AURA is benchmarked under the identical recipe against a plain 3D residual U-Net (**Baseline**) and a heavier multi-module variant (**Complex**), testing whether a single well-placed Transformer can match a more elaborate design.

**Deployment.** A FastAPI back end with a Niivue 3D front end: NIfTI upload, single-pass inference with per-region volumes and confidence, 3D rendering, and exportable report. The deployed pipeline reuses the evaluation inference path.

**Anticipated risks.** VRAM ceiling, fp16 attention instability, and severe class imbalance. Mitigations are baked in: attention only at the deepest stage, region-wise loss, tumor-centered sampling, and post-processing of small false-positive components.

## 4 Equipment Requirements

| **Item** | **Purpose** |
|---|---|
| Workstation GPU | NVIDIA RTX 40-series ($\geq$ 12 GB VRAM). |
| Workstation host | 32 GB RAM, 1 TB SSD. |
| Data storage | 2 TB SSD / NAS for BraTS NIfTI and `.npy` cache. |
| Software stack | PyTorch, MONAI, FastAPI, Niivue; CUDA 12.x. |

## 5 Expected Deliverables and Outcomes

**Deliverables.** BraTS pre-processing pipeline; a configurable backbone covering Baseline, AURA and Complex; unified evaluation with TTA, MC Dropout, post-processing and calibration; cross-model statistical comparison; FastAPI + Niivue web workstation; written report, poster, and oral presentation.

**Contributions.** Evidence on whether a single well-placed Transformer bottleneck matches a much heavier multi-module hybrid; an evaluation protocol that treats calibration and uncertainty as first-class metrics; an open-source clinical workstation as a deployment proof-of-concept.

**Training outcomes.** 3D deep learning architecture design, mixed-precision training, experimental design and statistical reporting, and full-stack development.

## 6 Schedule (Gantt Chart)

| Work Item \ Month | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Literature & dataset survey | | | | | | | | | | | | |
| Pre-processing pipeline | | | | | | | | | | | | |
| Baseline model | | | | | | | | | | | | |
| AURA implementation & train. | | | | | | | | | | | | |
| Complex variant & comparison | | | | | | | | | | | | |
| Calibration & uncertainty | | | | | | | | | | | | |
| Web workstation | | | | | | | | | | | | |
| Report, poster, presentation | | | | | | | | | | | | |
| **Cumulative (%)** | 8 | 18 | 28 | 40 | 50 | 60 | 70 | 80 | 88 | 94 | 97 | 100 |

## 7 References


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